

PAPER-CUT CHINESE GRAMMAR VIDEO

A Complete Production, Automation, Approval, QA, and Recovery Manual

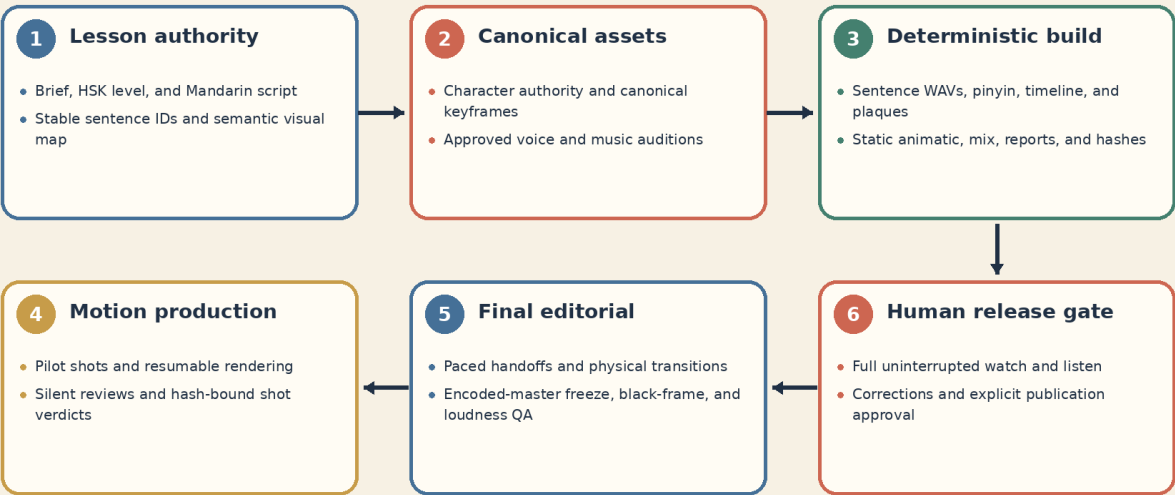
Built from the production experience of the 又……又…… and 我还以为……没想到…… HSK 3 lesson films

Prepared for Jonathan Caras

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Paper-Cut Chinese Grammar Video Production System

Generative tools propose. Deterministic systems establish authority. Humans approve.



Fail-closed rule: no missing file, stale hash, failed decode, pending visual verdict, or unapproved choice may enter final assembly.

Scope note: the corrected 又……又…… motion master was approved by Jonathan, published on his website, and shown to his Chinese teacher. The second case study has passed automated media checks and assistant visual qualification as a static audiovisual animatic, but it still awaits Jonathan's approval; motion generation remains blocked.

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1. Executive Summary

These videos are not produced by asking one generative model to “make a Chinese lesson video.” They are assembled through a gated production system. Generative tools are used where they are strongest—illustration, texture, environment invention, music ideation, and short motion performances. Deterministic tools control authoritative text, timing, compositing, transitions, manifests, and final assembly; count-sensitive generated imagery remains provisional until visually verified or rebuilt deterministically. Human approval is required wherever quality is subjective or where an automated pass cannot prove that the result feels right.

The central lesson from both projects is that technical validity is necessary but not sufficient. A file may decode perfectly, contain the exact frame count, pass a black-frame scan, and still feel slow, awkward, visually unstable, musically annoying, or typographically off-center. The pipeline therefore separates four ideas that are often confused: technical completion, visual correctness, editorial quality, and release approval.

State	What it proves	What it does not prove
Technical pass	The file exists, decodes, matches stream/frame contracts, and is hash-recorded.	That the image is attractive, the action is semantically correct, or the pacing feels good.
Visual pass	Sampled frames preserve identity, anatomy, props, style, and intended action.	That the complete film feels smooth when watched uninterrupted.
Editorial pass	Timing, handoffs, music, subtitles, and continuity feel coherent in the full sequence.	That the owner approves publication.
Release approval	The human owner watched/listened to the complete current master and explicitly approved it.	Nothing earlier may substitute for this decision.

A robust production is therefore “fail closed.” Missing clips, stale hashes, failed decodes, incomplete verdicts, unapproved voices, rejected music, or pending human review stop the pipeline. The system should never silently substitute a plausible asset or treat a wrapper process exit code as proof that the actual work succeeded.

2. What the Finished Product Is

The target is a short HSK-level Mandarin grammar lesson presented as a handcrafted paper theater. Recurring characters teach through examples, contrasts, practice, quizzes, and a closing learner prompt. The final spoken language may be Mandarin only, even when the source lesson is bilingual. Deterministic Simplified Chinese appears in a large subtitle plaque; a smaller tone-mark pinyin bar sits beneath it. Every grammar example is mapped to visible evidence. Jonathan’s current house music direction is smooth, happy, and legato; universally, the score must remain quiet beneath speech and resist fatigue. Tactile sound effects reinforce paper actions without becoming a beat.

2.1 Quality Definition

- Language is accurate, level-appropriate, and spoken naturally enough for learners.
- Every example visually communicates both sides of the grammar relationship—for example expectation versus reality, or two simultaneous predicates.
- Recurring characters maintain face, hair, clothing, proportions, and paper construction.
- Generated models never control authoritative Chinese, pinyin, titles, quiz answers, or timing.
- Ordinary dialogue does not contain unexplained multi-second pauses.
- Same-scene speaker changes are motivated physical paper replacements rather than unexplained visual snaps.
- Music supports concentration and is substantially quieter during speech.
- The encoded master—not only source artwork—passes subtitle, transition, black-frame, freeze, loudness, and decode QA.
- Publication remains blocked until the owner approves the complete current master.

3. Required Setup

3.1 Hardware

There are two viable architectures. A local-heavy system gives maximum control and lower marginal cost but requires substantial GPU memory. A hybrid system uses paid APIs for difficult image, speech, or video tasks and can run on a smaller local workstation.

Component	Practical recommendation	Why it matters
Operating system	Linux is the proven environment.	Local model stacks, CUDA tooling, FFmpeg, and automation scripts are easiest to reproduce there.
GPU for local still-image generation	Modern NVIDIA GPU with enough VRAM for the chosen FLUX/SDXL checkpoint.	Produces local style experiments and cleanup candidates.
GPU for the proven local H3 configuration	Observed in August 2026: the project's NF4 wrapper at 1280×720-class guides treated about 40 GB free VRAM as its launch floor; a 48 GB-class device was practical.	This is an empirical result for the recorded checkpoint, components, wrapper, and settings—not a universal requirement. Offloading, quantization, resolution, frames, and model versions can change it.
System memory	64 GB or more is comfortable for large contact sheets, frame caches, Whisper, and model orchestration.	Prevents CPU-side bottlenecks and swap thrashing.
Storage	At least 100 GB free per active project; more for retained raw attempts and frame caches.	Native renders, silent reviews, extracted JPEG/PNG frames, WAV stems, snapshots, and rejected attempts accumulate quickly.
CPU	Modern multi-core CPU.	FFmpeg encoding, Pillow rendering, LaMa inpainting, hashing, and archive creation use CPU heavily.
Monitoring	Access to GPU process/memory reporting and render queue state.	The production must wait for unrelated workloads instead of killing them.

3.2 Core Software

Tool	Role	Required or optional
Python 3	Manifests, orchestration, Pillow rendering, compositing, QA reports.	Required
FFmpeg / ffprobe	Conform audio, strip native model audio, mix stems, encode, decode-test, inspect streams, detect black/frozen frames, measure loudness.	Required
Pillow	Deterministic Chinese/pinyin plaques, titles, contact sheets, physical transitions, masked composites.	Required
pypinyin	Initial tone-mark pinyin generation.	Required when pinyin is included; manual adjudication still required
Whisper or equivalent ASR	TTS transcript screening and wordless-music screening.	Strongly recommended
Edge TTS	Free audition and production voice lane.	Recommended
xAI generation utility/API	Paid Eve/Leo voice auditions and, when explicitly approved, stronger scene/environment generation.	Required for Jonathan's free-versus-paid house audition; every paid call still requires authorization
FLUX/SDXL image generator	Optional local scene/keyframe generation and img2img experiments.	At least one still-image route is required overall; this local route is optional
SAM / segmentation	Extract stable character cutouts from a clean reference for deterministic compositing.	Optional but highly useful
IOPaint / LaMa	Narrow removal of pseudo-text from otherwise approved art.	Optional but useful
MiniMax-H3 or other first-frame-guided image-to-video model	Short semantic motion performances.	Required only for full-motion version
Word/LibreOffice	Open and update document fields; optional PDF conversion.	Optional for production, useful for documentation

3.3 API and Credential Hygiene

Paid APIs must be explicit choices. Voice auditions should label free and paid lanes clearly. xAI calls should be made only after the owner authorizes paid use. Credentials belong in environment variables or protected configuration—not in prompts, manifests, reports, screenshots, ZIPs, or production summaries. Logs may record model, operation, and output path, but never secret values.

3.4 Recommended Project Layout

Use an independent project root for every grammar lesson. Never overwrite a previous film to “save time.” Reuse scripts and approved character authorities, but keep timelines, hashes, approvals, caches, and selected motion clips project-specific.

Directory	Contents
images/	Character authority, generated environments, cleaned plates, canonical keyframes, cutouts.
audio/raw/	Provider-native sentence files and auditions.
audio/conformed/	Production WAVs, normally 48 kHz / 24-bit.
audio/sfx/ and audio/score/	Tactile effects, selected music master, extended score bed.
audio/mixes/	Narration, SFX, score, and final speech-first mix.
manifests/	Script, scene map, pinyin, audio, timeline, shots, cache state.
scripts/	Generation, conforming, rendering, QA, recovery, assembly.
approvals/	Verbatim decisions and hashes for voices, music, pilots, keyframes, masters.
evidence/	Technical reports, snapshots, launch receipts, final QA ledgers.
work/h3_raw/	Native motion outputs including provenance audio.
work/h3_selected/	Silent selected review clips.
validation/	Per-shot samples, contact sheets, overview pages, transition and master QA.
previews/	Static animatics, pilots, audition movies, review cuts.
masters/	Current authoritative motion masters and delivery encodes.
snapshots/	Pre-render file/hash authority snapshots.

3.5 Minimum Manifest and State Contracts

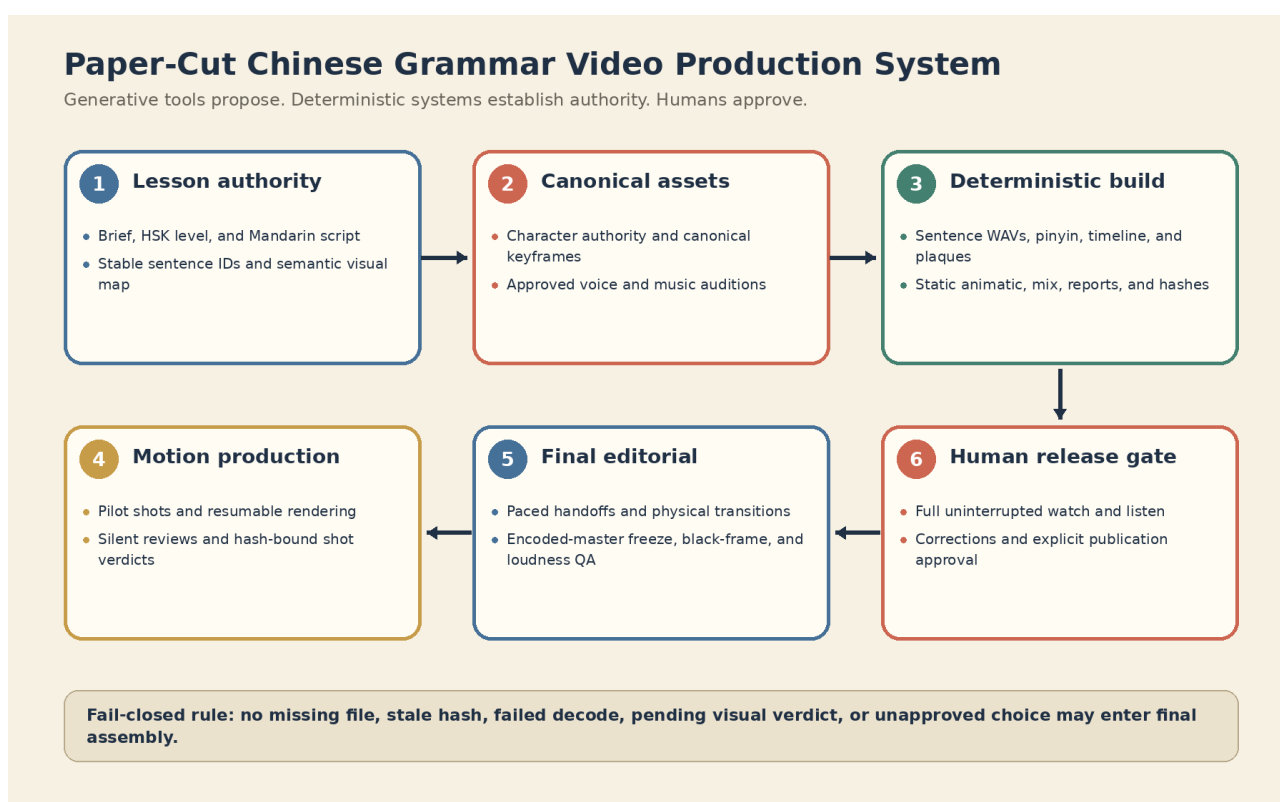
The folder tree becomes reproducible only when each promoted asset has a machine-readable record. Use UTF-8 JSON, a schema version, stable IDs, ISO timestamps with timezone, SHA-256 hashes, and explicit status transitions. Free-form prose may explain a verdict, but it must not replace the fields below.

Record	Minimum fields	Allowed progression
Project brief	schema_version, project_id, grammar_target, HSK level, spoken language, dimensions, FPS, approval flags.	DRAFT → BRIEF_APPROVED → SUPERSEDED
Sentence/TTS row	sentence_id, speaker, approved Chinese, provider/voice/rate, source and conformed paths, duration, ASR verdict, hashes.	DRAFT → GENERATED → REVIEW → PASS or REPAIR
Scene/keyframe card	scene_id, sentence_ids, teaching purpose, semantic evidence, people/prop counts, safe zone, forbidden content, source/output hashes.	DRAFT → GENERATED → CLEANUP or REGENERATE → PASS
Timeline event	event_id, sentence_id, scene_id, start/end frames, gap category, transition contract.	DRAFT → TIMELINE_LOCKED → SUPERSEDED

Record	Minimum fields	Allowed progression
Motion shot	shot_id, guide hash, prompt, model/settings, requested/returned frames, raw/silent paths, technical and visual status.	PENDING → RENDERED → TECH_PASS → VISUAL_PASS or REGENERATE
Approval record	gate name, exact asset hashes, reviewer, verbatim decision, timestamp, scope, superseded_by.	PENDING → APPROVED or REJECTED → SUPERSEDED
Master QA record	master hash, stream/frame/audio facts, anomaly classifications, sampled-review counts, human-review and publication flags.	BUILT → TECH_PASS → HUMAN_APPROVED → PUBLISHED

Any changed authority hash invalidates downstream approvals that depended on it. A repaired keyframe reopens its motion-shot verdicts; a changed sentence reopens the timeline, subtitles, mix, and master. The system should calculate invalidation rather than relying on memory.

4. System Architecture and Responsibility Boundaries



The system deliberately gives different responsibilities to different components. Generative models propose appearances and motion. Deterministic code establishes truth. Human reviewers decide taste, pedagogy, and release readiness.

Responsibility	Best owner	Reason
Chinese sentence authority	Approved script manifest	Generated text is unreliable and difficult to audit.
Pinyin authority	Local generation plus manual linguistic corrections	Automatic libraries do not reliably handle word grouping, neutral tones, or lesson-specific sandhi.
Mandarin linguistic review	Qualified Mandarin teacher, editor, or native-level reviewer	Grammar accuracy, naturalness, register, HSK suitability, and pedagogical pinyin require language expertise beyond file QA or product taste.
Character identity	Approved reference or exact extracted cutout	Full-frame generation drifts faces, clothing, and proportions.

Responsibility	Best owner	Reason
Environment invention	FLUX, xAI, or another image model	Generative systems are useful for style, props, and scene composition.
Object counts and quiz answers	Deterministic rebuild or strict visual gate	Image models frequently change counts.
Sentence timing	Measured conformed audio durations	Estimated reading times create dead air or clipping.
Transitions and titles	Local renderer	Exact duration, positioning, and repeatability are required.
Motion performance	First-frame-guided image-to-video model	Short local movement is a generative strength.
Final assembly	Fail-closed deterministic assembler	No missing or stale asset may slip through.
Taste and educational clarity	Human owner	No metric can prove “this feels slow,” “this music is annoying,” or “the image communicates the lesson.”

5. Approval Model: What Happens Automatically and What a Person Must Approve

The pipeline is designed to work autonomously between explicit gates. The human should not have to approve every command or every intermediate file; they should approve decisions that are expensive, subjective, or identity-defining.

Gate	Automatic work before gate	Human approval required	What approval does not include
Lesson brief	Convert source notes into a Mandarin script, stable IDs, scene map, pacing proposal.	Grammar scope, learner level, teaching logic, tone.	Voices, art, music, motion, final publication.
Language review	Prepare a sentence ledger, pinyin draft, and change report.	A qualified Mandarin reviewer checks grammar, naturalness, HSK level, tone-mark pinyin, and learner clarity.	Voice casting, art direction, motion, or publication.
Voice audition	Generate free Edge and paid xAI candidates on the same lines; level-match; decode and ASR-check.	Choose one provider/voice for each recurring character.	Acceptance of every generated sentence.
Canonical keyframes	Generate/clean environments, enforce counts, composite approved characters, build contact sheet.	Approve art direction, identity, semantics, and subtitle-safe composition.	Motion behavior or final edit.
Music audition	Generate distinct, level-matched wordless options; screen for vocal artifacts.	Choose musical direction and preferred track.	Final mix loudness.
Static audiovisual animatic	Conform all speech, build pinyin, timeline, subtitles, transitions, mix, encoded QA.	Approve lesson timing, voices in context, scene sequence, text hierarchy, music.	Full motion or publication.
Motion pilot	Render representative shots; technical and visual QA; produce contact sheets.	Approve motion language before scaling.	All remaining shots or final master.
Full-motion review	Render resumably, repair failures, assemble, run complete encoded QA.	Watch/listen uninterrupted and report editorial defects.	Publication unless explicitly stated.
Release	Make final corrections, new hash, delivery encode, archive/package.	Explicit approval to publish/upload.	Future revised versions.

5.1 The Rule for Approval Scope

Every approval should be recorded verbatim, dated, and bound to the asset hashes it actually covers. “The pilot is approved at this stage” means the pilot direction is approved; it does not approve later H3 shots, music, the full edit, or publication. A corrected master has a new hash and requires a new review decision.

5.2 Dependency Map and Parallel Work

The workflow is gated but not purely serial. After the script and qualified linguistic brief are approved, three lanes may proceed in parallel: scene/keyframe development, voice auditioning, and music-direction exploration. The measured timeline waits for approved sentence audio; the static animatic waits for approved keyframes, speech, pinyin, and mix. Expensive motion waits for the complete animatic gate.

Dependency	May proceed in parallel	Gate that unlocks the next stage
Approved script and linguistic brief	Scene cards/keyframes • voice auditions • music auditions	Approved sentence audio unlocks measured timing; approved scene cards unlock final keyframes.
Approved keyframes + speech + pinyin + mix	Deterministic animatic rendering and encoded QA	Jonathan’s contextual animatic approval unlocks motion pilots.
Approved representative motion pilots	Remaining shots may render resumably	Complete technical and hash-bound visual ledger unlocks assembly.
Assembled encoded master	Automatic media QA and sampled visual review	Jonathan’s uninterrupted full review unlocks release correction or publication.

Each phase must declare its inputs, outputs, automatic evidence, reviewer role, approval scope, and invalidation triggers. Work may be parallelized only when the missing dependency cannot silently change its authority later.

6. End-to-End Production Procedure

6.1 Phase 0 — Initialize the Independent Project

1. Create the directory structure before any paid generation so output writes cannot fail after an API request.
2. Copy only reusable authorities: character reference, approved SFX, selected music source when intentionally reused, and generic scripts.
3. Create project_brief.json with language, grammar, target level, target format, style constraints, characters, and approval state.
4. Set publication_authorized and motion_render_authorized to false.
5. Record source-script provenance and preserve the unedited source document.

Completion criterion: the project can be rebuilt without referring to mutable chat context, and no old project timeline, H3 cache, hash, approval, or selected clip is treated as authority.

6.2 Phase 1 — Adapt the Lesson Script

Convert the source lesson into short, natural Mandarin sentences suitable for the target HSK level. Keep each sentence complete and independently replaceable. Use stable IDs such as A01_F and A02_M. The script should teach the grammar through a progression: greeting, meaning, component explanation, examples, summary rule, learner quiz, model answer, and transfer prompt.

For every sentence, define the visible evidence. A sentence like “我还以为这家饭店很贵，没想到一点儿也不贵” needs an expensive expectation and an affordable reality—not merely two characters talking beside a restaurant.

6.2.1 Qualified Mandarin Review

Before voice production, a qualified Mandarin teacher, editor, or native-level reviewer should check grammar accuracy, naturalness, register, HSK suitability, word choice, punctuation, and whether the explanations teach the intended distinction. After deterministic pinyin is prepared, the same gate checks tone marks, word grouping, neutral tones, and sandhi. Record requested changes against stable sentence IDs. Jonathan retains product and release authority; the linguistic reviewer supplies language expertise rather than publication approval.

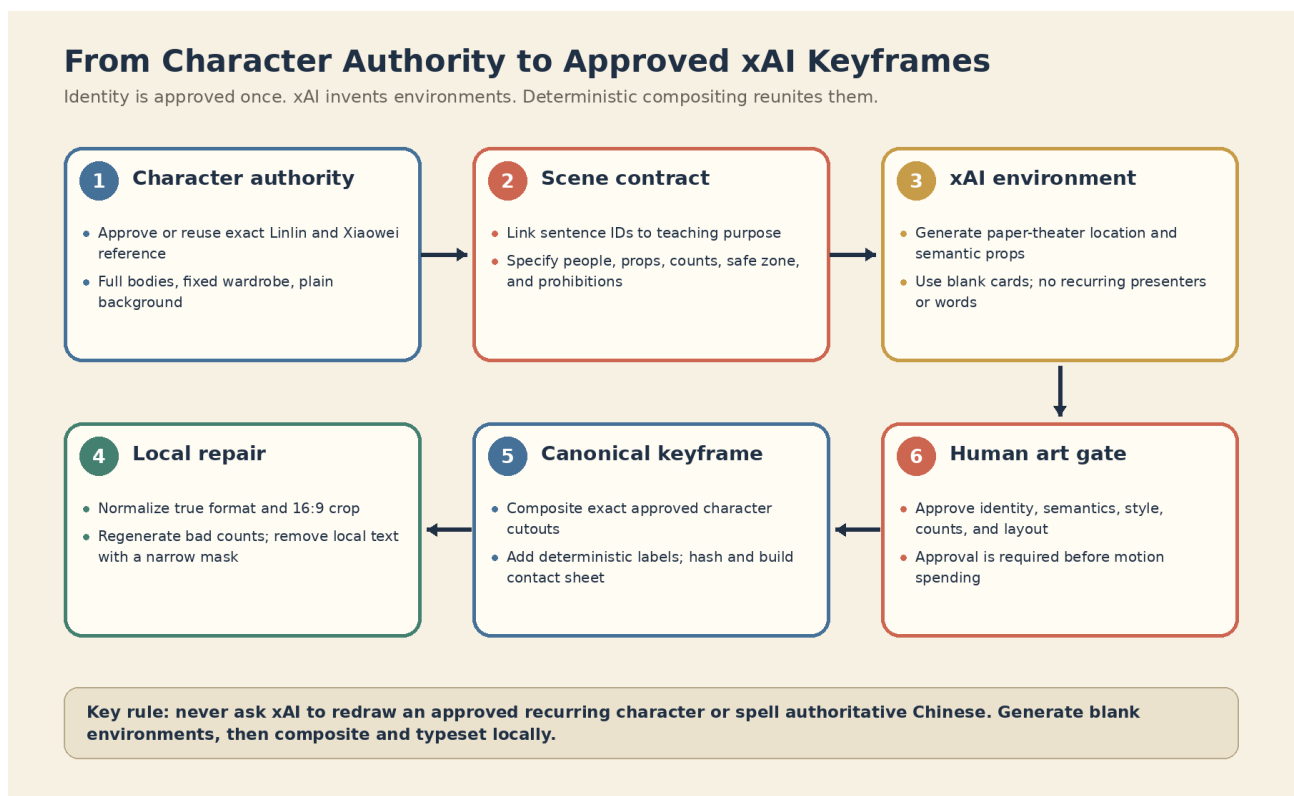
6.3 Phase 2 — Create the Semantic Scene Map

Each scene record should include its purpose, sentence IDs, expectation/reality or predicate mapping, required people and object counts, forbidden text, and lower subtitle-safe region. Generated images must not be trusted to supply labels; use blank cards and add authoritative labels later.

- Specify exact counts (“eight coins on the expectation side, two on the reality side”).
- Specify static regions (“wall icons remain pinned”).
- Specify the endpoint (“door remains open and characters settle”).
- Specify forbidden behavior (“no morph, no duplicate person, no pseudo-writing”).
- Keep important semantic props above the subtitle plaques.

6.4 Phase 3 — Character References and xAI Keyframe Creation

This is the bridge between the written lesson and the visual film. “Character reference” means the approved identity authority for recurring presenters such as Linlin and Xiaowei. “Keyframe” means the canonical still image that defines one scene before animation. The process should not ask xAI to reinvent both identity and scenery in every scene. Instead, identity is approved once, xAI is used primarily to invent the paper-theater environment and semantic props, and the exact recurring characters are composited locally.



6.4.1 Create or Reuse the Character Reference Authority

If the recurring characters already have an approved reference from an earlier film, copy that exact file into the new project and record its SHA-256. Do not regenerate them merely because a new lesson has started. If no authority exists, create reference candidates before scene generation: full-body Linlin and Xiaowei on a plain cream paper background, consistent front or three-quarter view, fixed clothing and colors, complete hands and shoes, neutral readable poses, and no scene-specific props or text.

1. Generate a small candidate set rather than a whole movie’s artwork.
2. Reject glossy 3D, photorealism, malformed anatomy, inconsistent proportions, or changing wardrobe.
3. Have the human approve the characters’ faces, hair, clothes, body proportions, age impression, warmth, and paper construction.
4. Promote one exact image as the identity authority; preserve rejected candidates separately.
5. Hash the promoted file and make every later keyframe point back to it.

Human gate: the recurring character reference requires explicit approval because facial identity and appeal are subjective and expensive to correct after motion generation. Automatic checks can prove file format, dimensions, and hash stability, but they cannot decide whether Linlin and Xiaowei “look right.”

6.4.2 Turn the Lesson into Chapter/Scene Reference Cards

Before calling xAI, create one machine-readable reference card for every lesson chapter or scene. This is more precise than a loose image prompt. Each card links the spoken lesson to visible evidence and becomes the QA contract for the future keyframe.

Field	What to record	Example
Scene identity	Stable scene/keyframe ID and sentence IDs.	KF05 • A11_M–A12_F
Teaching purpose	What the learner must understand from the picture.	Expected expensive restaurant versus surprisingly affordable reality.
Left/right semantic evidence	The two sides of the grammar relationship.	Eight gold tokens on the expectation side; two on the reality side.
Required people	Exact support-character count; whether recurring presenters will be composited later.	No presenters in the xAI plate; exactly one server only when the answer is revealed.
Required props/counts	All count-sensitive objects.	Exactly eight separate coins and exactly two separate coins.
Composition	Camera, central action band, presenter parking areas, subtitle-safe zone.	Front-facing wide paper theater; quiet lower 28 percent.
Forbidden content	Anything the model must not invent.	No words, pseudo-writing, currency symbols, logos, interface graphics, extra people.
Motion affordance	What later animation needs room to move.	A hinged reality card and an open path for a paper-overlay reveal.

Human gate: approve the scene-reference cards for pedagogical meaning before paid generation. The human is not approving final artwork yet; they are approving what each picture must communicate.

6.4.3 Prompt xAI for Environment Plates

Use xAI when explicitly authorized as the paid image-generation lane. The preferred prompt asks for a handcrafted miniature Chinese paper theater, tactile cut paper and felt, visible fibers, layered cardboard depth, soft physical shadows, a controlled warm palette, and a front-facing composition. Ask xAI for the environment and semantic props—not the recurring presenters and not authoritative labels.

The production contract is vendor-neutral: any still-image provider may replace xAI if it can follow the same scene card, preserve blank text regions, produce decodable assets, and pass the same count, semantic, style, and human approval gates. xAI is documented here as the proven paid implementation, not as the only valid provider.

A robust prompt contains six layers: (1) paper-theater style; (2) exact scene layout; (3) exact people and object counts; (4) negative constraints; (5) the lower subtitle-safe zone; and (6) space where approved presenter cutouts will later stand. Example: "Split restaurant price comparison, no presenters: left table beside exactly eight separate countable gold tokens; right meal beside exactly two separate tokens. No other coins, menus, writing, currency symbols, logos, or watermarks. Keep the bottom 28 percent as quiet cream paper stage."

Practical command pattern: `xai-gen.py image --prompt "SCENE CONTRACT EXPANDED INTO A PROMPT" --out images/xai_environments/KF05_environment.png`. The xAI utility may return JPEG bytes even when the requested name ends in .png, so inspect the real magic bytes, convert to actual PNG, crop the central action band to 16:9, resize to 1280×720, and retain the original response as provenance.

6.4.4 Technical QA and Machine-Assisted Visual Review

Automatic evidence is limited to what the software actually measures: file existence, full decode, magic bytes, color mode, dimensions, hash, and contact-sheet creation. People and object counts, pseudo-text, anatomy, semantic evidence, tactile style, composition, and subtitle-safe placement remain reviewer judgments unless a specific validated detector is documented for that field.

1. Verify that the output exists and fully decodes.
2. Detect the actual image format rather than trusting the extension.
3. Normalize dimensions and color mode.
4. Record the exact SHA-256 and generate a labeled overview/contact sheet.
5. Have a reviewer compare people, props, and exact counts against the scene-reference card.
6. Have a reviewer check pseudo-text, logos, anatomy, paper style, composition, safe zones, and both semantic predicates.
7. Record PASS, CLEANUP, or REGENERATE with the file hash, reviewer role, and exact defect.

6.4.5 Repair, Regenerate, or Reject

Choose the narrowest remedy. If exact counts or scene meaning are wrong, regenerate the environment with a stricter prompt. If one small unwanted word appears on an otherwise approved blank card, preserve the original and remove only that ink with a narrow LaMa/IOPaint mask. Never expose the entire approved plate to `img2img` merely to erase one label. After any repair, decode, hash, inspect the full frame and enlarged crop, and regenerate the contact sheet.

6.4.6 Composite the Approved Recurring Characters

Extract Linlin and Xiaowei from the clean reference using SAM or another segmentation method. Use tight full-body boxes, several foreground points on head/torso/legs, and background points outside the silhouette. Reject masks that include the stage, backboard, or props. The source with the simplest plain background is usually more important than changing segmentation models or seeds.

- 1. Save transparent cutouts with alpha and preserve rejected masks.
- 2. Inspect complete hair, hands, clothing, legs, and shoes.
- 3. Scale and place the exact cutouts scene by scene without covering semantic props.
- 4. Use a narrow cream torn-paper outline and subtle physical shadow only when needed for readability.
- 5. Keep recurring identity pixels stable; do not ask xAI to redraw the characters after approval.
- 6. Add Chinese titles, labels, calendar days, quiz text, and pinyin later through the deterministic renderer.

6.4.7 Promote Canonical Keyframes and Obtain Approval

The final keyframe is the cleaned xAI environment plus exact approved character cutouts and any deterministic non-dialogue labels needed for the animatic. Build one overview sheet containing every keyframe in lesson order. Review each frame for identity, people/object counts, anatomy, both sides of the grammar meaning, paper style, pseudo-text, composition, subtitle-safe space, and future motion affordance.

Result	Meaning	Next action
PASS	The exact hashed keyframe is canonical.	It may enter the static animatic and later become first-frame motion authority.
CLEANUP	The composition is approved but one narrow local defect remains.	Perform masked deterministic repair, then re-QA and create a new hash.
REGENERATE	Meaning, count, identity, or overall composition is wrong.	Preserve the failed candidate and produce a new environment plate.

Human gate: Jonathan approves the keyframe set as still artwork. This approves the characters, style, semantic mapping, counts, and subtitle-safe layout only. It does not approve motion, timing, voices, music, final edit, or publication.

6.5 Phase 4 — Audition Voices Before Bulk Generation

This became a mandatory gate after the second project exposed a process mistake: bulk Edge sentences were generated before Jonathan had compared free and paid voices. The files were retained only as provisional candidates; they were not declared final.

- 1. Choose one representative line per recurring character.
- 2. Generate the same text with a free voice lane (for example Edge Xiaoxiao/Yunxi) and a paid lane (for example xAI Eve/Leo).
- 3. Create the output directory before a paid call.
- 4. Conform all samples to the same channel format and loudness target.
- 5. Decode every file and run ASR.
- 6. Send clearly labeled samples.
- 7. Record the selected voice, rate, provider, user wording, file hashes, and approval time.
- 8. Generate or promote bulk speech only after selection.

In the second project, Jonathan selected both Edge voices: Xiaoxiao at -20% for Linlin and Yunxi at -15% for Xiaowei. The paid xAI auditions remained evidence only.

6.6 Phase 5 — Generate One Complete Sentence per File

Sentence-level assets are one of the highest-value architectural decisions. If a word is wrong, regenerate only that sentence. Do not create one long narration file that makes repairs destructive and complicates timing.

- 1. Generate the provider-native file.
- 2. Conform to a production WAV, typically 48 kHz / 24-bit.
- 3. Run a full decode.
- 4. Measure duration and record SHA-256.
- 5. Transcribe with ASR and compare normalized text.
- 6. Use a second ASR model or human listening for ambiguous lines.
- 7. Reject semantic substitutions, even when similarity remains high.
- 8. Update the sentence row to PASS only after adjudication.

Normalize Simplified/Traditional differences before judging ASR. Treat spoken homophones such as 她/他 correctly. Names such as 琳琳 may be recognized as 玲玲 even when pronunciation is correct. ASR is evidence—not the final listener.

6.7 Phase 6 — Create Deterministic Chinese and Pinyin

Chinese subtitles come directly from the approved script. The house pinyin policy is word-based Hanyu Pinyin spacing with tone marks, manually adjudicated neutral tones, and reviewed 一/不 tone changes. Maintain a lesson exception dictionary for pedagogical display. Examples of intended grouping include yìqǐ, yí jù, yìdiǎnr, míngbai, jīngshen, rènào, dōngxì, wǒmen, xuéxí, zhōngwén, yuánlái, fāxiàn, nuǎnhuo, and yǒuyìsi; a qualified Mandarin reviewer must approve the actual lesson output rather than treating this list as a substitute for review.

Use at most two Chinese lines and two pinyin lines. At 1280×720, establish project-specific minimum sizes in the brief; a proven starting floor is about 32 px for Chinese and 19 px for pinyin. Keep plaques within horizontal and vertical safe margins, preserve strong text/background contrast, and test at actual phone and website playback size. If text does not fit, wrap at approved phrase boundaries, shorten the approved copy, or use a scene-specific layout—never shrink without a minimum-size gate. Convert unsupported full-width punctuation in the Latin pinyin font to reviewed ASCII equivalents. Verify the longest and widest lines after encoding.

6.7.1 Visible-Ink Centering

Font origins and advance widths are not the visible shape of the glyphs. CJK fonts contain large top bearings; long lines and punctuation have asymmetric side bearings. Center the visible ink bounding box in both axes and center the plaque geometry independently around the frame midpoint.

Reference formula: $\text{bbox} = \text{font.getbbox}(\text{text})$; $\text{ink_width} = \text{bbox}[2] - \text{bbox}[0]$; $\text{text_x} = (\text{frame_width} - \text{ink_width})/2 - \text{bbox}[0]$; $\text{text_y} = (\text{plaque_top} + \text{plaque_bottom} - (\text{bbox}[1] + \text{bbox}[3]))/2$. Use symmetric plaque padding around the ink bounds.

6.8 Phase 7 — Build the Content-Derived Timeline

Do not force a round runtime. Measure the conformed speech and build around it. The second film reached 146.041667 seconds—inside the requested 2–2.5 minutes—because its 115.176 seconds of speech were combined with intentional breaths, transitions, a quiz hold, and a closing hold. No padding was added merely to hit 150 seconds.

Timing element	Recommended starting value	QA rule
First speech after scene arrival	0.35–0.50 s	Enough to register the new scene, not enough to feel idle.
Ordinary speaker handoff	0.70–0.90 s	Any unexplained ordinary gap over 1.2 s is a defect until justified.
Post-sentence scene breath	0.80–1.20 s	Longer requires visible action or pedagogical reason.
Scene transition	0.25–0.50 s	Overlap existing time; do not extend runtime by stacking transitions after pauses.
Quiz thinking hold	2.0–2.5 s	Intentional and labeled distinctly from ordinary gaps.
Final learner response hold	About 4 s	May remain longer, but the paper stage should still feel alive.

6.9 Phase 8 — Music and Sound Design

Music must be auditioned by musical direction, not by making three nearly identical gain or tempo variants. The first project proved that slowing a rhythmic ostinato does not remove the sensation of “beep beep beat.” When Jonathan rejected all rhythmic options, the direction changed to smooth jazz: sustained piano/Rhodes/guitar harmony, major-sixth or major-ninth color, rising melody, sparse bass, and brush swells without a strong backbeat.

1. Generate three musically distinct, level-matched excerpts.
2. Screen for accidental vocal-like artifacts with ASR.
3. Ask the human to choose based on the isolated music.
4. Rebuild the full score from the exact selected master.
5. Mark previous score approvals superseded.
6. Mix music quieter beneath speech and apply sidechain ducking.
7. Use tactile SFX only for meaningful paper or semantic events.

A proven starting mix uses music near 0.40 linear gain before ducking, a relatively strong sidechain ratio, fast attack, and roughly 400–500 ms release. These values are starting points, not universal mastering rules. The final mix needs loudness measurement and human listening.

6.10 Phase 9 — Build and Approve the Static Audiovisual Animatic

The static animatic is the cheapest place to discover expensive problems. It combines final selected voices, real timing, deterministic subtitles, pinyin, keyframes, title placement, transitions, music, SFX, and the intended quiz holds. It should be fully encoded and QA'd, not merely played from an editor timeline.

1. Render every frame deterministically at the delivery resolution and FPS.
2. Mux the approved mix.
3. Run a full decode and ffprobe.
4. Detect black frames and frozen runs.
5. Extract start/end samples for every utterance.
6. Extract before/mid/after samples for every transition.
7. Inspect title, closing, longest subtitle, quiz question, hold, and answer.
8. Send the complete animatic to the human for art, voice, pacing, music, subtitle, and lesson-logic approval.

Motion generation should remain blocked until this gate passes. In the second case study, production voices were already selected and the static animatic passed automatic media checks plus assistant visual qualification; the complete animatic still awaits Jonathan's contextual approval of artwork, voice performance, timing, music, subtitles, and lesson flow. Motion generation and publication remain unauthorized.

6.11 Phase 10 — Design Motion as Independent Shots

Each motion clip has a stable shot ID, canonical first frame, prompt, seed, requested frames, dimensions, FPS, native output, silent review output, report path, and pilot flag. Prompts must specify both motion and stillness. "Preserve the first frame" is too weak; say which person moves, which props remain fixed, the exact count, and the desired endpoint.

6.12 Phase 11 — Pilot Before Scale

Choose representative difficult shots: a character/prop interaction, energetic movement, environmental motion, and a quiz answer. Render only these first. Strip native model audio and treat it as provenance. Run complete technical and visual QA. Human approval of the pilot is the authorization to spend GPU time on the remaining batch—not approval of the future final master.

6.13 Phase 12 — Render Resumably

A long batch must survive interruption. A clip may be reused only if raw and silent files exist, the report says complete, hashes match, and the silent review fully decodes. A wrapper process exiting 0 is not proof that child renders finished. Inventory the actual files, report completed/failed/pending counts, and preserve partial progress.

Before launch, create a pre-render snapshot of guides, manifests, scripts, approvals, model components, processor files, and selected audio. Verify the snapshot hash manifest. Inspect active GPU work and wait rather than terminating unrelated jobs. Use the model project's qualified virtual environment; system Python may lack dependencies even when the model stack is correctly installed.

6.14 Phase 13 — Full Technical and Visual Shot QA

QA type	Automatic checks	Human/visual checks
Technical	Existence, decode, one video stream, no silent-review audio, codec, dimensions, FPS, exact frame count, duration, first-frame fidelity, hashes.	None of these prove semantic correctness.
Identity/anatomy	Contact sheets are generated automatically.	Face, hair, clothes, body proportions, hands, limbs, duplication.
Semantics	Prompt/object-count contract is loaded.	Both grammar concepts are visibly communicated.
Static discipline	Pixel comparison can prove guarded regions unchanged after deterministic repair.	Background motion must feel intentional; endpoint must be holdable.
Style/text	OCR may supplement review.	Paper texture, no pseudo-text, no glossy drift or morphing.

Define frame-0 fidelity numerically. Exact pixel equality is meaningful on lossless pre-encode frames and deterministic composites. Lossy encoded comparisons must use documented tolerances—such as maximum channel error plus SSIM or PSNR—while accounting for color conversion and chroma subsampling. Bind the guide, decoded frame, metric, tolerance, and hashes in the technical report.

Visual verdicts should be bound to exact video SHA-256 values. If a selected clip changes, its verdict returns to pending. Final assembly accepts only a complete PASS_ALL_N ledger.

6.15 Phase 14 — Repair Without Thrashing

After the first failure, simplify the action and tighten constraints. After a second miss, pivot instead of cycling seeds. Preserve rejected attempts and evidence. A powerful deterministic repair is to freeze the approved background to frame 0 while retaining only valid H3 character/prop regions through a feathered mask. Prove unchanged pixels outside the guard numerically, then inspect the composite edges and final pose.

6.16 Phase 15 — Fail-Closed Assembly

The assembler refuses to run unless the expected clip count is present, every technical test passes, the visual ledger says PASS_ALL_N, each row is hash-matched, the approved mix exists, and frame/runtime contracts are known. Extracted-frame caches must be invalidated whenever a source hash changes; otherwise a corrected clip can appear unchanged because stale frames were reused.

6.17 Phase 16 — Editorial Pacing and Handoff Repair

Do not play a short generative clip at native speed and then freeze its last frame for several seconds. That creates the exact defect reported in the first film: speech, dead hold, slight image snap, next speech. Map the complete motion monotonically across its editorial window, never reverse frames, and use a short rigid paper overlay when switching between independently encoded same-scene takes.

Freeze detection creates a review queue; it is not a goal to force the detector to report zero intervals. Classify each detected hold as intentional pedagogical stillness, an acceptable endpoint hold, or an editorial defect. A one-pixel stepped paper-stage settle at 1280 px is only 0.078% of frame width and may be used when it genuinely reads as tactile paper life, but never add meaningless movement solely to defeat `freezedetect`. Any settle still requires human viewing for jitter and edge tearing.

6.18 Phase 17 — Final Encoded Master QA

The project brief—not this manual—sets the release contract: canvas, FPS, pixel format, video codec/profile, audio sample rate/channels, integrated-loudness target and tolerance, true-peak ceiling in dBTP, and delivery-size limit. Peak sample level in dBFS is not interchangeable with measured true peak in dBTP. Starting values such as 0.40 music gain or 10:1 ducking describe one case study, not a universal standard.

- Complete FFmpeg decode.
- Exact frame count, FPS, dimensions, duration, and audio format.
- Integrated loudness, loudness range, and true peak.
- Black-frame scan.
- Freeze scan with every interval reviewed.
- All sentence start/end samples.
- All same-scene handoffs before/mid/after.
- All scene transitions before/mid/after.
- Opening and closing titles.
- Longest Chinese and pinyin lines.
- Quiz question, thinking hold, answer, and final response hold.
- SHA-256 and byte size.
- Full uninterrupted human watch/listen.

6.19 Phase 18 — Delivery and Archiving

Keep a high-quality review master and make a separate size-capped delivery encode. Use two-pass H.264 when a platform requires a file under a specific limit; verify actual bytes and full decode. For audio delivery, include the individual sentence WAVs, selected music source/master, score bed, used SFX, timed stems, final mix, and exact extracted AAC track. Exclude rejected assets unless requested. If the archive is too large, produce independently openable ZIPs with their own README and manifest—not opaque binary fragments—and run archive integrity tests.

7. Automatic QA Versus Human QA

The best system automates everything measurable and reserves human attention for perception, meaning, and taste. Neither side can replace the other.

Area	Automatically QA'd	Human must QA
Script	Stable IDs, sentence/scene references, counts, missing fields.	Naturalness, HSK appropriateness, pedagogy, whether examples are useful.
TTS	Decode, duration, sample rate, peak/RMS, hash, ASR similarity, second-model transcript.	Voice preference, warmth, pace, prosody, subtle mispronunciation, character fit.
Pinyin	Initial generation, line width, missing glyph detection through encoded samples.	Word grouping, neutral tones, sandhi, pedagogical conventions.
Artwork	Dimensions, format/magic, hash, and contact-sheet production.	People/object counts, identity, anatomy, style, exact visual meaning, attractiveness, and whether the scene feels coherent.
Motion	Decode, frames, FPS, dimensions, audio stripping, frame-0 fidelity, hashes, sample extraction.	Mutation, wrong action, hands, prop behavior, paper style, endpoint quality.
Audio mix	Loudness, peak, decode, sample rate/channels, ASR wordless screen.	Music taste, speech intelligibility, fatigue, whether ducking feels natural.
Timing	Gap computation, labeled quiz holds, freeze detection.	Whether the lesson feels rushed, slow, awkward, or engaging.
Subtitles	Visible-ink geometry, frame-safe bounds, plaque symmetry, encoded samples.	Readability at real viewing size and whether the hierarchy feels right.
Master	Full decode, stream contracts, black/freeze scans, sample grids, hashes.	Uninterrupted editorial watch and final release decision.

7.1 Why Sampled QA Is Not a Full Watch

Contact sheets are excellent at finding identity, anatomy, object-count, subtitle, and transition defects. They do not reproduce time. A freeze can occur between samples; a pause can feel much longer than its numeric duration; music can become tiring through repetition; a slight snap can be obvious only during continuous playback. Therefore sampled QA qualifies a master for human review—it does not replace that review.

8. Detailed Failure Case Studies and Lessons

8.1 Wrong TTS Word: 打散房间 Instead of 打扫房间

In the first film, one narration file said 打散房间 rather than 打扫房间. Because every sentence was independent, only A18_M had to be regenerated. Lesson: sentence-level assets turn semantic repair into a narrow operation. ASR plus listening must check meaning, not only audio validity.

8.2 Voice Audition Was Initially Skipped

In the second film, free Edge files were generated before the required free-versus-paid audition. Jonathan corrected the process. The generated files were reclassified as provisional, paid xAI Eve/Leo samples were created on the same lines, all samples were level-matched and ASR-checked, and Jonathan selected both Edge voices. Lesson: a technically valid voice is not an approved voice. Audition before bulk production.

8.3 Paid Call Reached the API but Could Not Save

The first xAI audition calls reached the service but failed at local file creation because the output directory did not exist. The directory was created and the calls were rerun. Lesson: create and verify destination directories before paid requests; local packaging failures can waste paid generations.

8.4 Yunxi Pronounced 句型 as 剧情

Two ASR models consistently heard 剧情 where the script required 句型. Rather than accepting the error or endlessly retrying the same voice, the HSK-appropriate sentence was rewritten as “就试试这样说吧.” The new file passed the second

model exactly. Lesson: when a provider systematically mishandles one phrase, a natural equivalent can be safer than brittle phonetic hacks.

8.5 Image Bytes Did Not Match the Extension

FFmpeg reported an invalid PNG signature because some files had JPEG bytes under .png names. The guides were normalized to true PNG. Lesson: validate image magic bytes and normalize formats before batching; filenames are not proof of content type.

8.6 Full-Frame Art Generation Drifted to 3D Dolls

The second film's first 12-keyframe FLUX batch completed technically but failed visually: smooth 3D-doll style replaced tactile paper, pseudo-writing appeared, support characters were omitted, and expectation/reality staging weakened. The batch was rejected. The method pivoted to xAI-generated paper environments with no recurring presenters, followed by deterministic compositing of exact approved character cutouts. Lesson: separate scene invention from identity authority.

8.7 SAM Segmented the Stage Instead of the Characters

The first SAM extraction attempts used a title/teaching stage whose cardboard backboards were connected to the characters. The masks included scenery. Those cutouts were preserved as rejected evidence. SAM was then pointed at the clean character reference with a plain cream background, tight boxes, foreground torso/leg points, and background points. The resulting alpha masks isolated the full bodies. Lesson: segmentation quality depends more on the source authority and prompts than on rerunning the same model.

8.8 Generated Counts and Pseudo-Text Were Wrong

xAI produced useful environment plates but one restaurant count and one speech-bubble count were wrong; a restaurant answer plate included English words. The count scenes were regenerated with explicit, separately countable objects. The words were removed through a narrow LaMa mask rather than whole-frame regeneration. Lesson: count and text failures are expected; use exact contracts, narrow repair, and re-QA.

8.9 Semantic Objects Fell Under the Subtitle Safe Area

Coins and the closing card path occupied the lower region where Chinese and pinyin plaques would appear. Those layers were shifted upward deterministically and the lower stage filled with sampled paper color. Lesson: artwork is not approved until viewed with the actual subtitle geometry, not just as a clean image.

8.10 Subtitle Plaque Was Raised but Text Still Sat Too Low

In the first film, raising the plaque fixed frame-edge safety but did not solve the real complaint: the visible Chinese ink sat low inside the plaque, touching the bottom and leaving excess top space. The cause was the CJK font's invisible top bearing. Visible-glyph bounding boxes fixed vertical centering.

8.11 Long Closing Subtitles Were Horizontally Off-Center

Advance-width centering and asymmetric plaque padding looked acceptable on short lines but failed on the long closing sentence. Both Chinese and pinyin were changed to visible-ink horizontal centering, and plaque padding became symmetric around x=640. The encoded closing frame—not merely source rendering—was measured and approved. Lesson: always test the longest line and punctuation-heavy line.

8.12 Pinyin Had Missing-Glyph Squares

The second animatic initially used full-width Chinese punctuation and quotation marks with a Latin pinyin font. Encoded samples showed missing-glyph boxes. The pinyin punctuation was converted to supported ASCII, and common word grouping was manually polished. Lesson: a Unicode string can be linguistically correct but typographically unsupported; inspect encoded pixels.

8.13 Title Covered Character Faces/Hair

A deterministic title plaque was initially centered too low. Smoke-test frames revealed it crossed the presenters. The plaque was reduced and moved to the top paper header. Lesson: title QA must include face/hair clearance and final encoded framing.

8.14 FFmpeg Duration Syntax Failed

One FFmpeg build rejected leading-dot duration syntax such as .02, and the proposed fade-out had no computed start time. The unnecessary fade was removed rather than hiding the error. Lesson: validate command portability on a small asset; use explicit 0.02 syntax and calculate time-dependent filters.

8.15 Music Stayed Annoying After Being Slowed

The first response to “too loud and too fast” was to create slower, softer versions of the same rhythmic track. Jonathan rejected all of them because the underlying beep-like ostinato remained. The production then created genuinely different smooth-jazz performances, later made happier through harmony and melodic direction rather than a stronger beat. Lesson: respond to the perceptual cause, not only the numeric parameter.

8.16 Wrapper Exit 0 Did Not Prove Batch Completion

The H3 wrapper completed with exit code 0, but output was truncated and the wrapper alone could not prove all shot artifacts were present. A file-based 27-shot technical harness established completion through existence, decode, dimensions, FPS, frame contracts, hashes, and contact sheets. Lesson: process status is orchestration evidence; artifact evidence proves production completion.

8.17 S13A Wall Icons Fell to the Floor

Full visual QA passed 26 of 27 first-pass clips. In S13A, wall icons detached and fell, reversing the intended pinned-background behavior. A tighter prompt and new seed were attempted. The stable solution was deterministic: freeze every pixel outside guarded Linlin/Xiaowei regions to frame 0 while retaining the valid H3 gestures. Exact maximum difference 0 was proven on the lossless pre-encode composite outside the masks; the encoded result then received normal decode and visual QA. Lesson: use generative motion where it is valid and deterministic compositing where it is not.

8.18 The Complete Film Passed Automation but Felt Slow and Awkward

The original first-film master passed 27/27 shot QA, 29 scene samples, 14 transitions, black-frame detection, subtitle QA, and loudness checks. Jonathan’s uninterrupted watch still found frequent pauses followed by small image jumps. The root cause was structural: 5–7-second dialogue gaps and H3 clips that played at native speed, froze, then snapped to the next independently encoded take.

The correction reduced runtime from 218.0 to 113.416667 seconds while preserving all 25 sentences, 48 spoken 叉 tokens, three quiz holds, and the closing response hold. Regular handoffs became 0.8 seconds, same-scene changes became 0.3-second rigid paper overlays, full H3 performances were mapped monotonically across speaking windows, and a one-pixel stage settle eliminated residual ≥ 0.5 -second freezes. Lesson: the owner’s full watch outranks a green dashboard.

9. QA Ledger: Recommended Automatic Checks

Stage	Required checks	Fail condition
Source art	Decode, actual format, dimensions, hash.	Wrong bytes/extension, missing source, wrong canvas.
Sentence audio	Decode, 48 kHz target, duration, peak/RMS, ASR, hash.	Semantic mismatch, clipping, unreadable file, pending adjudication.
Keyframe	Exact scene ID, dimensions, no authoritative generated text, contact-sheet verdict.	Identity drift, count error, unsafe subtitle area, pseudo-text.
Motion raw	Existence, decode, requested frames, streams, dimensions, FPS.	Missing/partial file, wrong frame count, decode error.
Silent review	No audio, full decode, hash, frame-0 fidelity.	Native model audio leaks or guide mismatch.
Visual verdict	PASS/REGENERATE notes and exact SHA-256.	Pending row, stale hash, unresolved defect.
Timeline	Every event duration, gap categories, runtime at exact frame boundary.	Overlap, clipping, unexplained gap > threshold.
Mix	Decode, sample rate/channels, loudness, peak, ASR wordless music screen.	Speech masked, clipping, wrong streams.
Encoded master	Full decode, frames, FPS, dimensions, duration, audio, black/freeze scans, contact sheets, hash.	Any unreviewed anomaly or contract mismatch.
Delivery	Actual byte size, decode, archive integrity, manifest count/hashes.	Size cap exceeded, corrupt ZIP, missing files.

10. Human Review Checklist

10.1 Qualified Language Review

- Is the grammar explanation accurate and natural in Mandarin?
- Is every sentence appropriate for the stated HSK level and learner age?
- Do examples express the intended contrast without introducing a different grammar pattern?
- Are tone-mark pinyin, word grouping, neutral tones, and sandhi correct?
- Are teacher-requested changes recorded against stable sentence IDs and propagated downstream?

10.2 Voice Audition

- Does the voice fit the character?
- Is the speed slow enough for HSK 3 without sounding unnatural?
- Are tone, warmth, and energy suitable for a lesson?
- Does either provider mispronounce the grammar target or recurring words?
- Choose each recurring character explicitly.

10.3 Keyframe Approval

- Do Linlin and Xiaowei look like the approved characters?
- Does every example clearly show both halves of the grammar idea?
- Are exact counts correct?
- Are support characters present and not duplicated?
- Is the paper style tactile rather than glossy 3D?
- Is the lower subtitle area clear?
- Is any generated pseudo-text visible?

10.4 Static Animatic Approval

- Does the lesson feel neither rushed nor padded?
- Are ordinary gaps short and quiz holds intentional?
- Are Chinese and pinyin readable on the actual device?
- Does the music feel smooth, happy, and quiet enough?
- Do transitions feel physical and motivated?
- Does the title cover faces or compete with subtitles?
- Does the quiz provide enough time to think?

10.5 Motion Pilot Approval

- Does the motion feel like paper theater rather than morphing video?
- Do hands, faces, props, and object counts stay stable?
- Does the moving action teach the intended meaning?
- Do backgrounds remain fixed where required?
- Would this motion language be acceptable across the whole film?

10.6 Full Master Approval

- Watch from start to finish without skipping.
- Listen on the expected device and at normal volume.
- Note any place attention drops, everything pauses, or a scene snaps.
- Check recurring music fatigue.
- Read the longest subtitle and final subtitle.
- Confirm the learner prompt has a usable response hold.
- Approve publication explicitly only after corrections are incorporated into the current hashed master.

11. Reproducibility, Provenance, and Recovery

Every promoted asset should have a recorded source, output path, model/tool, settings, hash, and verdict. Rejected attempts are preserved outside the production authority set. This is not bureaucracy: it makes interrupted batches resumable, prevents stale caches, proves which file was approved, and allows narrow repairs without recreating the film.

- Use SHA-256 for guides, speech, selected motion, mixes, masters, snapshots, and delivery ZIPs.
- Bind visual verdicts to exact clip hashes.
- Invalidate frame caches when source hashes change.
- Record exact model checkpoint and interpreter environment.
- Create pre-render snapshots before long GPU batches.
- Preserve native model audio only as provenance; use silent reviews in editorial.
- Never overwrite the sole original or rejected evidence.

Recovery procedure: (1) inventory authority manifests and actual files; (2) verify hashes and decodes; (3) classify every item as complete, failed, pending, or stale; (4) invalidate dependent caches and approvals; (5) restore from the latest verified snapshot when an authority is missing or corrupt; (6) resume only pending work; and (7) regenerate the relevant gate report before assembly. Never infer the recovery state from a wrapper process alone.

Retention should be deliberate rather than infinite. Until final approval, keep every authority, selected asset, failed attempt needed to explain or reproduce a decision, and the latest verified snapshot. After publication, retain manifests, hashes, approvals, masters, source authorities, repair evidence, and representative failures; large redundant rejected renders may be pruned only under a documented policy after confirming that no recovery or audit path depends on them.

12. Cost-Aware Production Options

Mode	Recommended stack	Trade-off
Mostly local / low-cost	Edge TTS, the required limited xAI voice audition, local FLUX, local music model, local H3, FFmpeg/Pillow/Whisper.	Low marginal production cost after the audition; highest hardware and operations burden.
Hybrid	Free Edge voices when approved; xAI for difficult environment plates; local deterministic compositing and H3.	Good balance of quality, cost, and control.
API-heavy	Paid TTS/image/video APIs plus local renderer/QA.	Lower hardware requirement; higher recurring cost and provider variability.

Record a per-project spend plan before paid generation: authorized provider lanes, estimated calls, maximum retries, stop threshold, expected local GPU hours, expected storage, and who may approve additional spend. Paid-call preflight must confirm that the destination exists and is writable, credentials work, the request is logged idempotently, and budget remains. A successful API response with a failed local save still counts as a production failure and may still incur cost.

Regardless of mode, keep deterministic text, timing, subtitles, hashes, and final assembly local. Paid quality does not remove the need for QA.

13. Practical “Start Here” Build Plan

1. Install and verify Python, FFmpeg/ffprobe, Pillow, pypinyin, Whisper, and at least one TTS route.
2. Choose an image route and, if doing local motion, qualify the GPU/model environment before the project starts.
3. Create the standard folder tree and a project_brief.json.
4. Adapt the script into stable sentence IDs and a semantic scene map.
5. Obtain qualified Mandarin review of grammar, naturalness, HSK level, and later the final pinyin.
6. Run free-versus-paid voice auditions; obtain explicit selections.
7. Generate and QA all sentence WAVs.
8. Create environment plates and an approved character authority; composite deterministically where possible.
9. Generate and manually correct pinyin.
10. Build a content-derived timeline and speech-first mix.
11. Render and human-approve the complete static animatic.
12. Design four representative motion pilots; render, QA, and obtain pilot approval.
13. Snapshot authorities; run the resumable full motion batch.
14. Complete technical and visual shot QA; repair failures using the two-miss pivot rule.
15. Assemble fail closed; run full encoded QA.
16. Have the owner watch uninterrupted; correct pacing and visual issues.
17. Produce the release master, size-capped delivery copy, audio package, manifests, and hashes.
18. Publish only after explicit approval.

14. Case Study Status at Time of Writing

Project	Current verified state	Human gate
又……又……	Complete motion film corrected from 218.0 s to 113.416667 s. 25 sentences, 27/27 motion shots, no ≥ 0.5 s freeze runs, 12/12 same-scene handoffs, 14/14 transitions, centered Chinese/pinyin.	Approved by Jonathan, published on his website, and shown to his Chinese teacher.
我还以为……没想到……	146.041667 s static audiovisual animatic. Production voices were selected at audition; 23/23 Edge sentence files, 13/13 canonical/composited keyframes, 12/12 transitions, clean black-frame scan, freeze scan reviewed, -16.3 LUFS / -1.4 dBFS measured peak.	Automated media checks and assistant visual qualification passed. Awaiting Jonathan's contextual approval of artwork, voice performance, timing, music, subtitles, and lesson flow before motion; publication is not authorized.

15. Master Verification Checklist

- Lesson brief and semantic map complete.
- Qualified Mandarin review completed and requested changes propagated.
- Voice audition includes free and paid lanes; selections recorded.
- Every sentence file decodes, matches language authority, and is hash-recorded.
- Chinese and pinyin are deterministic and visually centered.
- Keyframes preserve identity, counts, style, and subtitle-safe composition.
- Music is selected by the human and dialogue remains dominant.
- Static animatic is watched and approved before motion.
- Motion pilots pass technical/visual QA and receive human approval.
- Full batch is proven by files/reports/hashes—not wrapper status.
- Every selected clip has a current hash-bound visual PASS.
- No unresolved ordinary gap above 1.2 s.
- No unreviewed frozen run of 0.5 s or longer.
- Every handoff and transition passes encoded sample review.
- Black-frame scan is clean.
- Complete master decodes and matches stream/frame/audio contracts.
- Owner watches the complete current master and explicitly approves publication.

Appendix A — Reusable Command Patterns

The following commands are patterns. Paths, runtimes, and bitrates must be derived from the current project.

A.1 Decode Test

```
ffmpeg -v error -i INPUT.mp4 -f null -
```

A.2 Probe Streams and Frames

```
ffprobe -v error -count_frames -show_entries  
stream=index,codec_type,codec_name,width,height,r_frame_rate,nb_read_frames,sample_rate,channels:format=duration,size -of json INPUT.mp4
```

A.3 Black-Frame Detection

```
ffmpeg -hide_banner -nostats -i MASTER.mp4 -vf "blackdetect=d=0.08:pix_th=0.02" -an -f null -
```

A.4 Freeze Detection

```
ffmpeg -hide_banner -nostats -i MASTER.mp4 -vf "freezedetect=n=-50dB:d=0.5" -an -f null -
```

A.5 Loudness and True Peak

```
ffmpeg -hide_banner -nostats -i INPUT -map 0:a:0 -af ebur128=peak=true -f null -
```

A.6 Exact Audio Extraction

```
ffmpeg -y -v error -i DELIVERY.mp4 -map 0:a:0 -c:a copy exact_video_audio.m4a
```

A.7 Hash Verification

```
sha256sum FILE • sha256sum -c MANIFEST.sha256
```

A.8 ZIP Integrity

```
unzip -t PACKAGE.zip
```

Appendix B — Operating Principles

1. Generate broadly; approve narrowly.
2. Use generative models for proposals, deterministic tools for authority.
3. Measure real speech before timing.
4. Never stretch content to an arbitrary runtime.
5. Every grammar relationship must be visible.
6. One sentence per file.
7. Audition voices and music before scaling.
8. Approve canonical art before motion.
9. Pilot before full batch.
10. Bind every verdict to a hash.
11. Wrapper status is not artifact proof.
12. After two misses, pivot to deterministic repair.
13. Sampled QA qualifies; full human viewing decides.
14. A human-reported defect reopens the gate even when automation is green.
15. Publication is always a separate explicit approval.